

Introduction

I am currently a student at the University of Turku, though I'll be done soon. In the meantime I've gone deep on AI and built up a strong sense for what it takes to write code that actually holds up. First and foremost I'm a problem solver - give me something interesting and I'll run with it on my own until it's done right.

Education

2024–present	Master of Science , University of Turku <i>Major:</i> Software Engineering <i>Minor:</i> Tomorrow's AI
2020–2024	Bachelor of Science , University of Turku <i>Major:</i> Information and Communication Technology <i>Minor:</i> Mathematics, Optimization <i>Relevant courses:</i> • Web and mobile programming
2016–2019	High School , Tampereen Teknillinen Lukio <i>Mathematics and IT-focused Matek-line</i>

Publications

- **BSc Thesis:** [Polunetsintäalgoritmien esittely ja tehokkuusvertailu](#)
This thesis examines pathfinding algorithms from a computing-oriented point of view. It is a literature review, so it is mostly based on already published research. It also contains a small set of performance measurements of certain algorithms in a test environment. The thesis introduces the reader to a few pathfinding algorithms, their use cases, and compares their performance.

Relevant Work Experience

2025–present	AI Engineer , Cloop.io, Turku, Finland The first developer in this company, responsible for designing and building many parts of the current Cloop product, including: RAG (initial version, first vector database, current source scoring system), email system. Active on every part of the stack: Frontend (Chat component, product dashboard), Backend and Database. Relevant technologies: Python (FastAPI, Pydantic), PostgreSQL (incl. pgvector for RAG similarity search), TypeScript.
2024–present	Project member, Software Engineering & AI , VITAR project, Turku, Finland A frozen startup project where we attempted to develop an AI research assistant for biolab workflows. In the beginning I was the only software developer involved in this project. My tasks involved mobile development (Android Studio, Java), software architecture design, and managing AI systems (my first introduction to RAG, Vector databases, N8N and many many more).

- 2023 **Project member**, Information Technology Labs, University of Turku
In this project me and a few other students take part in developing a checkout application to be used with digital learning platforms as a SaaS-service. We used TypeScript and the T3 Stack. My job was to build the backend of the webhook integration.
- 2020 **Summer trainee**, HiDucator Ltd, Turku, Finland
My main job was to edit educational videos for this e-learning company. I also helped in shooting and producing the raw material for video production.

Selected Projects

- **learn3fruit** [\[repo\]](#)
2025 to present Python TensorFlow TensorFlow.js A-Frame containerization JavaScript
This application aims to prove that a Computer Vision Neural Network can be used to power a WebAR/Mobile-AR application. The finished app will guide the user through the preparation of a fruit salad given using a built in recipe (step-by-step format), where the neural network knows where to display the 3D-Augmentations by locating the ingredients & tools (object detection). This is a part of my MSc Thesis. It currently exists in two git repos, a complete rewrite of what has been developed thus far is in the works.
- **CV management projects** [\[repo\]](#)
2025 to present LaTeX Node.js React JavaScript GNU Make
A group of git repos that aim to add version control as well as ‘.tex’, ‘.pdf’ and web renderings of my CV and portfolio.
- **imagerminer/imapp** (*archived*) [\[repo\]](#)
2024 C quirc React Native Android Studio Android NDK
Our entry for the 2024 Boost Digital sustainability hackathon. The aim was to come up with a more efficient way to read QR codes on mobile. Our plan was to port over & compile the quirc C-library for Android using Android’s Native Development Kit (NDK), and then embed this in a QR code reading application developed using React Native. This project was never finished, due to a 48h hackathon time limit.
- **MedMate** [\[repo\]](#)
2024 JavaScript React Native
Our concept app for a medication calendar. Developed for the University of Turku course "Lean Digital Business Design". Navigation and UI designs implemented, no real functionality.
- **graph-tests-2** [\[repo\]](#)
2023–2024 C++ Boost Boost Graph Library R GNU Make
Implementations and studying of different pathfinding algorithms. Part of my BSc Thesis Work.

Languages

Hungarian	Native
Finnish	Matriculation exam: native (L)
English	Matriculation exam: advanced (M)
Swedish	Matriculation exam: intermediate (M)

Programming Languages

- **Python** (*Professional*) – **libraries**: NumPy, Pandas, PyTorch, TensorFlow, FastAPI, asyncpg, Pydantic; **package/environment management**: pip, uv, pyenv, venv, jupyter

- **JavaScript** (*Proficient*) – **frameworks**: React, React Native, Vue 3, Node.js; **libraries**: Transformers.js
- **C & C++** (*Familiar*) – **libraries**: Boost, Boost Graph Library
- **Java** (*Familiar*)

Databases

- **SQL** (*Proficient*) – **flavors**: PostgreSQL, MariaDB
- **Vector databases** (*Proficient*) – **types**: Qdrant, pgvector for Postgres
- **MongoDB** (*Familiar*)

Tools & Platforms

- **Linux** (*Proficient*) – **distros**: Fedora, Ubuntu, Debian; **package managers**: dnf, apt
- **Command Line Tools** (*Proficient*) – **Examples**: bash, git, make, ssh
- **Containerization tools** (*Proficient*) – **environments**: Docker, Docker Compose, Podman
- **AI Tools** (*Proficient*) – **libraries**: PyTorch, TensorFlow; **environments**: Hugging Face; **LLMs & code generators**: ChatGPT, Claude Code